

 1M1B • AI FOR SUSTAINABILITY

EcoStream Climate AI

Multimodal AI for Climate Action • Translating Complex Atmospheric Telemetry into Highly Engaging Community Content



Climate Intelligence Engine



Community Action Platform

The Data Disconnect: Inaccessible Reality

Despite billions invested in smart city sensors, the resulting climate telemetry is overly technical, deeply siloed, and profoundly unengaging for everyday citizens. This creates a severe gap in ground-level awareness and civic action.

80%

Buried in PDFs

Municipal environmental records and crucial telemetry remain locked within static, inaccessible dashboards and bureaucratic documents.

<2%

Public Access Rate

Local residents rarely consult government portals. The data is presented using overly complex and dry scientific terminology.

5-7d

Manual Turnaround

The time required by underfunded civic creative teams to manually parse data and produce a single localized awareness campaign.

Global Alignment: **UN SDGs**

13 **Climate Action**

TARGET 13.3 — PRIMARY FOCUS

We ensure rapid climate education, public awareness-raising, and human capacity building. EcoStream effectively transforms raw, continuous atmospheric telemetry into actionable digital narratives that foster unparalleled community-level resilience and accountability.

11 **Sustainable Cities**

SECONDARY MUNICIPAL FOCUS

EcoStream empowers sustainable and inclusive communities by actively tracking and communicating ward-level Air Quality Indices (AQI), microclimate heat islands, and municipal waste indicators to accelerate local civic interventions.

Our Core Objective

“

How might we use advanced AI to translate complex, dry climate data into engaging multimedia campaigns so communities can become more sustainable and highly informed?

”

🌐 THE ECOSTREAM PROBLEM STATEMENT

Who We Are **Designing For**



Municipalities & NGOs

Civic bodies need to publish verified, localized climate telemetry quickly, without relying on massive creative departments or expensive video production agencies.



Content Creators

Creators require accurate storyboards, rapid fact-checking cues, and structured narrative hooks to scale viral, impactful climate literacy across social media.



Residents & Citizens

Everyday people demand bite-sized, contextual, and hyper-local insights that highlight clear daily actions to safeguard their family's health and environment.

Proposed AI Architecture

EcoStream acts as an automated content engine and a retrieval-augmented advisor, working together seamlessly.

🔧 1. Agentic Workflows

Autonomous agents continuously ingest streaming sensor telemetry and atmospheric alerts.

🧠 2. IBM Granite Models

Synthesizes dense climate metrics into 30-second conversational voiceover scripts and visual hooks.

📋 3. RAG Policy Grounding

Retrieval-Augmented Generation grounds all messaging in verified local municipal laws, actively preventing greenwashing.



End-to-End Processing Pipeline



Interactive **User Experience**

Conversational Frontend

An NGO coordinator requests an immediate air quality update for their specific ward.

User / Coordinator:

"Analyze this week's air quality telemetry for Hyderabad."

EcoStream Agent:

"AQI is up 15%. Here are 3 quick actions to cut commuter emissions, complete with storyboards."

Behind The Scenes

The autonomous process executing in milliseconds:

- ✓ **Input Parsed:** Ward ID and geographic bounds identified; intent extracted.
- ✓ **Model Queries:** Automated agents fetch streaming real-time sensor data.
- ✓ **RAG Lookup:** Vector index matches city clean air action guidelines.
- ✓ **Asset Gen:** Drafts script and visual hooks, applying highlight effects onto NGO logos.

Telemetry to **Ready Content**

Automated Creative Packaging

Data Input:

"Analyze Hyderabad air quality (AQI 165, PM2.5 elevated)."

Generated Voiceover:

"Notice the haze today? AQI is up 15%. Here are three immediate actions to reduce commuter emissions in our ward..."

Visual Direction Cue:

Split-screen graphic comparing single-occupant vehicular traffic with rapid electric metro transit lanes. Audio set to upbeat rhythm.



Responsible **AI Safeguards**

Ethical checks and balances are built directly into the core architecture.



Fairness

Actively monitors sensors equally across lower-income and affluent neighborhoods to prevent environmental geographic bias.



Transparency

Automatically embeds verifiable municipal citations and exact telemetry timestamps directly onto every generated video asset.



Anti-Greenwash

Strict system guardrails eliminate misleading statistics, rejecting exaggerated corporate claims and enforcing verified figures.



Data Privacy

Exclusively processes open-source public telemetry; absolutely zero personal user identity data is tracked or stored.

Measurable **Impact & Efficiency**

Turnaround Speed (Hrs → Mins)

95% Faster

Audience Retention Uplift

3.5x Uplift

Policy Citation Accuracy

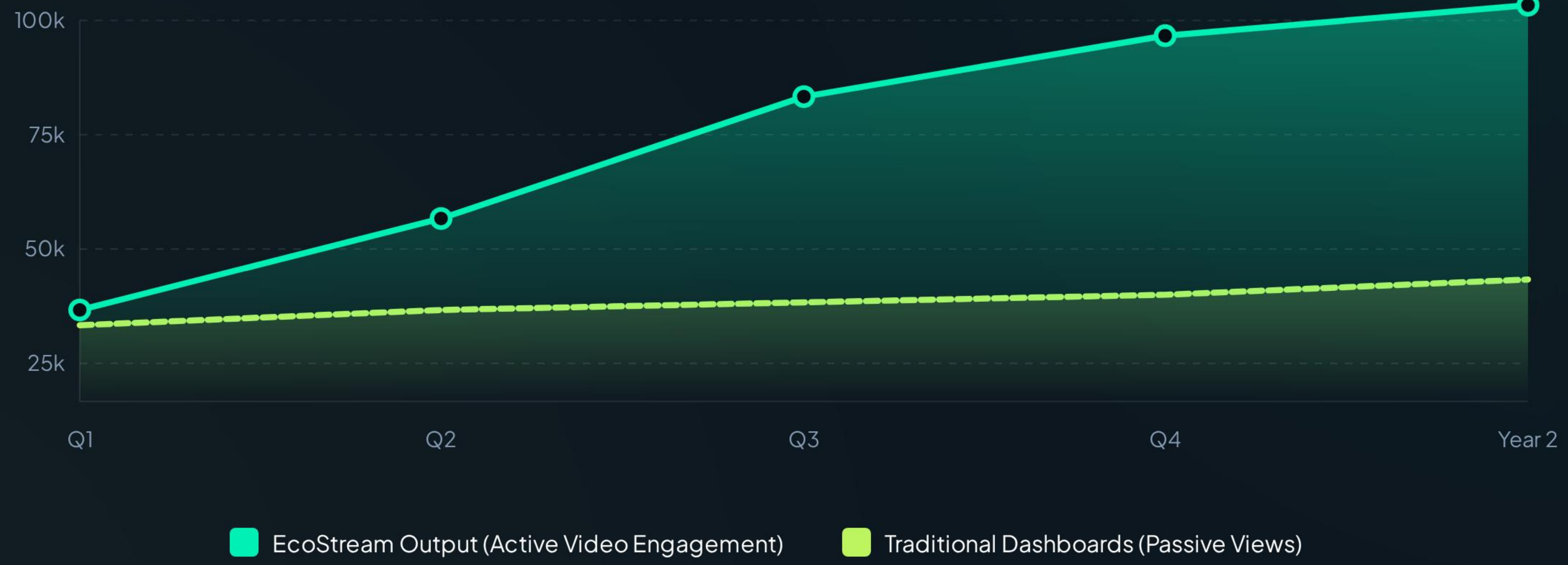
99.9% Grounded

Production Cost Reduction

88% Saved

EcoStream shifts climate communication from sluggish multi-day manual workflows to an instant, verifiable, community-first intelligence ecosystem.

Projected **Community Engagement** Trend



 THE FUTURE OF CLIMATE ENGAGEMENT

AI as a Force for **Sustainability**

EcoStream demonstrates that AI's greatest contribution to climate resilience lies not just in calculation, but in making complex scientific data human-centric, engaging, and directly actionable.



Think Critically



Apply AI Responsibly



Design with Purpose



Build for Impact

Image Sources



smart city municipal building with green technology and solar panels



content creator making a video about climate change in a modern studio



diverse community of citizens engaging in sustainable neighborhood gardening or clean energy



futuristic AI data center with green glowing lights and environmental monitoring screens



modern smart city street with electric public transit and digital billboards showing air quality

Image Sources



https://scwcontent.affino.com/AcuCustom/Sitename/DAM/042/Green_Energy_City_Minecraft_E_IT_Innoenergy_smart_cities_PR_rt.jpg

Source: www.smartcitiesworld.net



https://helios-i.mashable.com/imager/videos/02cUPPL8rWh6OZYko36hPq6/hero-image.fill.size_1248x702.v1783006006.png

Source: mashable.com



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Source: learningmole.com



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Source: stockcake.com



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